

Tobias Wegel

PHD STUDENT IN STATISTICAL MACHINE LEARNING

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Research Interests

Learning theory (aggregation and multi-objective learning), high-dimensional and game-theoretic statistics, optimization.

Education

ETH Zürich

Zürich, Switzerland

PHD IN COMPUTER SCIENCE, expected graduation in summer 2028.

2024 - present

Member of the Statistical Machine Learning Group and affiliated with ETH AI Center, advised by Prof. Fanny Yang.

MSC IN DATA SCIENCE, graduated with distinction (GPA 5.85, Swiss scale).

2021 - 2023

Georg-August-University Göttingen

Göttingen, Germany

BSC IN MATHEMATICAL DATA SCIENCE, graduated with distinction (GPA 1.1, German scale).

2018 - 2021

BSC IN PURE MATHEMATICS, graduated with distinction (GPA 1.1, German scale).

2018 - 2021

Professional and Research Experience

ETH Zürich

Zürich, Switzerland

(HEAD) TEACHING ASSISTANT, Introduction to ML and Guarantees for ML.

2024 - present

Simons Institute for the Theory of Computing at UC Berkeley

Berkeley, CA, United States

VISITING GRADUATE STUDENT, part of the Modern Paradigms in Generalization program.

Sep. 2024 - Dec. 2024

University of Oxford

Oxford, United Kingdom

VISITING RESEARCHER, with Prof. Patrick Rebeschini at the Department of Statistics.

Mar. 2023 - Jan. 2024

Neurocat GmbH

Berlin, Germany

MACHINE LEARNING RESEARCHER, research on certifiable robustness of AI models.

Aug. 2020 - Sep. 2021

Publications and Preprints

Reconciling Universal and Uniform Learning with Q-Aggregation

Submission, 2026

Aggregation with Exponential Weights Is Optimal in Expectation

Submission, 2026

Time-sensitive anytime-valid testing

Submission, 2026

Hedging on the Frontier: Learning New Tasks with Few Samples

ICML 2026 spotlight

Cutting LLM Evaluation Costs with SySRs: A Bandit Algorithm that Provably Exploits Model Similarity

ICML 2026

Sharp risk bounds for early-stopping in Gaussian linear regression

AISTATS 2026

On the sample complexity of semi-supervised multi-objective learning

NeurIPS 2025 spotlight

Learning Pareto manifolds in high dimensions: How can regularization help?

AISTATS 2025

Selected Talks

Feb. 2026 **OxCSML Seminar**, Theoretical aspects of learning with multiple objectives

University of Oxford

Mar. 2025 **Theory Seminar**, Early-stopping for linear regression over convex bodies

ETH AI Center

Skills and Languages

Technical: Python (NumPy, pandas, PyTorch), C, SQL, Git, Bash, LaTeX

Languages: German (native), English (fluent)